

Effect of degradation on the structural behaviour of an existing cantilever reinforced concrete bridge in Southern Italy

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Abstract

The performance of cantilever reinforced concrete bridges is fundamental topic in assessment and retrofit of Italian infrastructures, because the static scheme of the so-called “Gerber bridges” was very popular between 50’s and 70’s of XX Century. Many of them present severe damage due to the drainage of the deck waters inside the joint and to the consequent degradation of Gerber saddles. Furthermore, the state of corrosion of the reinforcements together with the actual degree of freedom of internal constraints, often partially blocked due to degradation and subsequent maintenance over time, modify the structural behaviour and performance both in terms of service and ultimate limit states. This paper presents a case study of a reinforced concrete bridge with a three-span cantilever scheme, designed in southern Italy by Riccardo Morandi, in which this problem led to a difficult assessment of the actual performance of the bridge. The on-site investigations and the bridge modelling in its current condition, taking into account the effects of degradation in the Gerber saddles, made it possible to evaluate the state of the bridge after more than 60 years from construction.

1 Case-study bridge

The bridge over the Salso river in Licata (Sicily, Southern Italy), was designed by Riccardo Morandi; it is a three-span bridge with a cantilever girder scheme (also called Gerber bridge), with side spans of 33.10 m and a central span, constituted by the two cantilevers and an isostatic beam, with a total length of 49.60 m. The overall width of the deck is 19.00 m, while the total length is 115.80 m. The original drawing with the 3D view is shown in Figure 1.

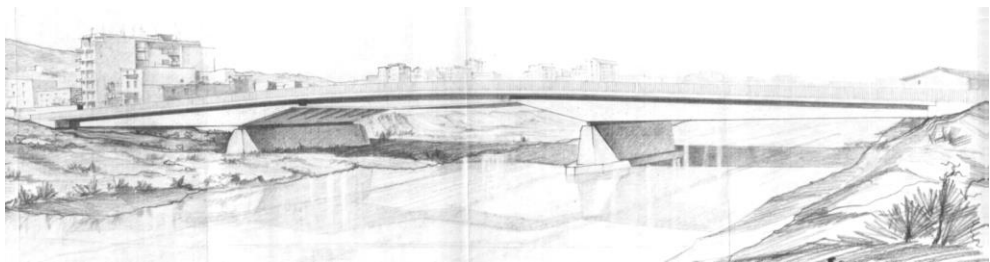


Fig. 1 View of the bridge taken from the original drawings.

In the two side spans and in the cantilevers, the deck is composed of eight side-by-side asymmetrical T-type beams made of reinforced concrete with a web thickness of 400 mm and a variable height from 1.40 m to 2.50 m. The upper slab is 200 mm thick. The girder is completed by three transverse beams for each span and by a bottom slab near the piers, with a thickness varying from 220 to 300 mm. The midspan is made up of eight prestressed concrete beams with variable height and double T section, completed by a reinforced concrete slab with a thickness of 200 mm and three transverse beams. The central beams are prestressed through the Morandi’s patent of M5 type tendons. The values of prestressing forces were obtained from the original documents. Materials are: concrete for side spans and cantilever with $f_{ck} = 25$ MPa; concrete for prestressed girder with $f_{ck} = 45$ MPa; ordinary reinforcement steel type Aq50, with yielding strength $f_{yk} = 270$ MPa; prestressing steel with tensile failure strength $f_{ptk} = 1700$ MPa.

Information about the bridge's state of conservation was gathered through visual inspections, in-situ investigations, tests on materials and structural elements, to achieve the adequate level of knowledge of the structure required by the Italian Code and by the recent Guidelines for Existing Bridge Assessment [1]. In this way it was possible to develop Finite Element structural models that would allow reliable evaluation of the current state of the bridge both in terms of stress and strain states (Service and Ultimate Limit States).

2 Safety assessment of cantilever and Gerber saddle

Figure 2 shows the geometry of the cantilever and the arrangement of the reinforcement bars. The lower corbel of the saddle was considered the weak element because the upper corbel of the central beam is prestressed, with a greater capacity guaranteed by the presence of prestressing anchorages and the related reinforcements.

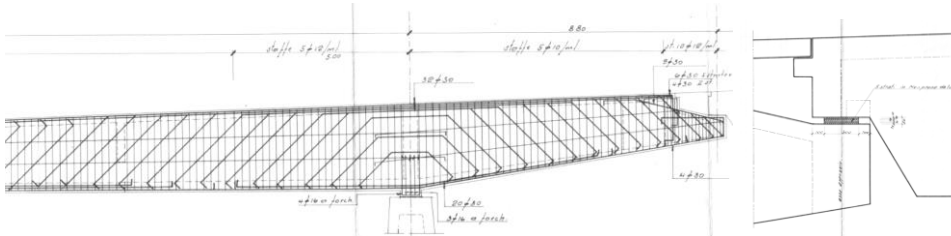


Fig. 2 Arrangement of cantilever reinforcements and geometry of the Gerber saddle.

For assessment of the capacity of the Gerber saddle, the provisions of Eurocode 2 were initially considered, using the strut-and-tie models provided in section 10.9.4.6 of EC2 [2]. Since, in the present case the problem is that of checking an existing saddle and not to design a new half-joint, the schemes supplied by the codes cannot be applied immediately. The particular geometry of the saddle designed by Morandi, which presents a soft connection between beam cross-section and corbel, instead of the classic dapped-end beam, leads to idealized strut-and-tie patterns that differ too greatly from the actual arrangement of the saddle. Hence, the strut-and-tie schemes supplied by the codes cannot represent the actual behaviour of the saddle.

Actually, from the original drawings, it is evident that the saddle was not poorly dimensioned at the time of the project, showing no state of suffering or cracks, apart degradation (Fig. 3).



Fig. 3 Pictures of the brige. General view and Gerber saddle.

The recent English recommendations for checking existing half-joints [3] as well as the interesting studies of Desnerck et al. [4] give useful indications on the behaviour of Gerber saddles in old bridges, also taking degradation into account, but they all refer to classical schemes of two dapped-end beams with a short half-joint.

In order to perform a reliable check on the Gerber saddle in this case-study, a 3D model made of shell elements was developed, following the actual geometry (Fig. 4) and considering the upper slab, the transverse beam within the saddle, and the lower flange of the longitudinal beam. In this way it is possible to take into account the actual strength contributions and the trend of the tensile and compressive principal stresses inside the structural element.

By operating a longitudinal section passing through the bearing of the central beam, it is possible to establish the tendency of the isostatic curves of tension and compression, in order to build a strut-and-tie model that is closer to the actual element. Comparing the trend of the principal stresses with the arrangement of the reinforcements conceived by Morandi (Fig. 2), it is possible to see how the designer well adapted reinforcements to the actual behaviour of the element; hence a new strut-and-tie model has to be conceived, taking into account the actual arrangement of reinforcements and the location of the struts. The model specifically adapted to the case-study is shown in Figure 4.

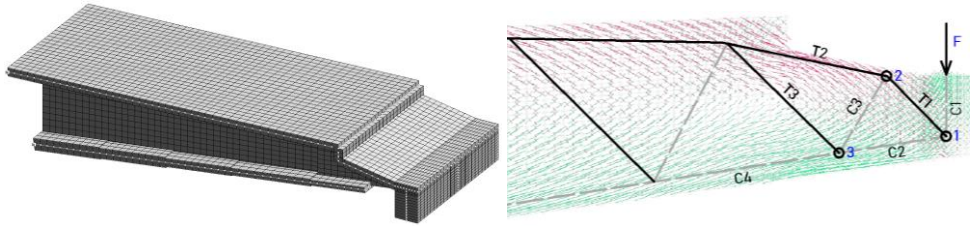


Fig. 4 3D Finite Element Model of cantilever and Gerber saddle. Strut-and-tie model of Gerber saddle specifically adopted in the case-study.

The result obtained for the check at ULS, using partial safety coefficients of loads and materials provided by Eurocode, is that the weakest element is tie T₂, with a capacity $F_{max} = 1195$ kN and a demand $V_{Ed} = 1189$ kN, bringing the safety coefficient equal to 1.005, bordering on acceptability.

Considering that the application of partial safety coefficients at the ULS cannot give an effective measure of the global safety coefficient at saddle failure, the check was repeated, albeit in a simplistic way, taking all the partial safety coefficients to 1. In this way a direct check was carried out when failure occurred: the estimated capacity is $F_{max} = 1370$ kN with a demand $V_{Ed} = 873$ kN, bringing the actual global safety coefficient equal to 1.57, which can be considered satisfactory.

The demand considered here is found through the moving loads of Eurocode 1 and not with the original ones of the design time. Only for research purposes, the authors repeated the same calculations applying the moving loads provided by the 1962 Italian code (the code applied for the design), finding the unexpected result of a greater demand than the current code: $V_{Ed,1962} = 1021$ kN, bringing the actual global safety coefficient to 1.34. However, the result is that the Gerber saddle in the original state shows a good safety factor.

3 Degradation of Gerber saddle

The deterioration of the materials due to ageing and the corrosion of the most exposed reinforcements modify the previous results, especially by considering the environment in which the bridge is placed: the mouth of a river, towards the sea. It is essential to know the actual state of reinforcements, and what level of corrosion could lead to the sudden failure of the saddle, due to degradation of the weakest element. From the previous evaluations, in intact conditions the weakest element of the truss in the proposed model is tie T₂. The area A_s of the tie, corresponding to the tensile stressed reinforcement, was considered totally effective in the previous evaluation; the initial corrosion-free value of this reinforcement will be named A_0 . Then, for a first approach, the effects of a corrosion process is considered limited to reduction of the area of reinforcements due to widespread corrosion induced by drainage of water inside the expansion joint and the saddle. Hence, the reduced area value of the corroded reinforcements A_{cor} which leads to a unitary global safety factor, could be evaluated, considering this condition as the one of the saddle failure. The corrosion rate CR is given by the following relation:

$$CR = 1 - \frac{A_{cor}}{A_0} = 1 - \frac{SF}{SF_0} = 1 - \frac{1}{SF} \quad (1)$$

where SF is the safety factor in the undamaged condition. With $SF = 1.57$ (moving loads of Eurocode 1), the result is $CR = 36\%$ while with $SF = 1.34$ (moving loads of Italian 1962 code), the result is $CR = 25\%$, which means that a 25% mass loss for corrosion would lead to failure. The result is comforting since no corrosion of this magnitude was found through in-situ investigations.

This evaluation is carried out in the simple hypothesis in which corrosion affects only the reduction of the area of the reinforcement and not concrete strength, bond or steel strength.

From Figure 3 it can be noted that the outermost beam presents the most deteriorated half-joint, with damage due to water drainage and corrosion of exposed reinforcements in the bottom corbel. A maintenance intervention closed the damaged zone with thixotropic mortar, but the deterioration of the saddle did not stop, and the phenomenon is still advancing with corrosion of reinforcements and drainage of water from the joint that induces damage of internal beams, too.

However, in the present case the bond of the smoothed rebar is ensured by the end hook, and concrete cracking causes reduction of the concrete compressive strength of the ties only. Thus, the following concomitant effects could be realistically assumed:

- a decrease of effective reinforcement area of ties and stirrups for mass loss, due to corrosion;
- a decrease of the concrete strength for struts.

According to the results of in-situ inspections a reduced reinforcement area of $0.85 A_0$ will be considered, and an average strength of concrete $f_{cd}' = 0.80 f_{cd}$, as suggested in literature studies [4] and ACI318 recommendations [5]. With these assumptions, the weakest element is the corroded tie T_2 in Figure 4, and the check on the saddle supplies a minimum safety factor $SF = 1.14$, fully acceptable, since the struts are always the elements with greatest capacity, even with a reduction in the strength of concrete. This confirms the robustness of the saddle designed by Morandi, both in terms of reinforcements (ties) and concrete struts, considering the possibility of redistribution offered by the transverse beam.

4 Conclusions

From the analyses carried out on the case-study bridge it emerges that the Gerber saddle has to be studied through models specifically conceived, because the schemes supplied by Codes often cannot be applied for existing saddles with the actual geometry and reinforcement arrangements. The results of the safety assessment show that the Gerber saddle of the case-study bridge is a fairly robust element, so that the likelihood of local failure is reduced compared to the condition of suffering of the most stressed cross-section of the cantilever on the pier top. This seems to indicate a global collapse mechanism as being more likely than the local one of the half-joint, at least in static conditions for vertical loads, even for the worst-case scenario of saddle damage.

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